| High energy molecule | Total number formed during TCA | No. of ATP generated per molecule | Total number of ATPs |
|---|---|---|---|
| NADH | 3 | 3 | 9 |
| $FADH_2$ | 1 | 2 | 2 |
| GTP | 1 | 1 | 1 |
| Total no. of ATP per acetyl coA molecule | | | 12 |

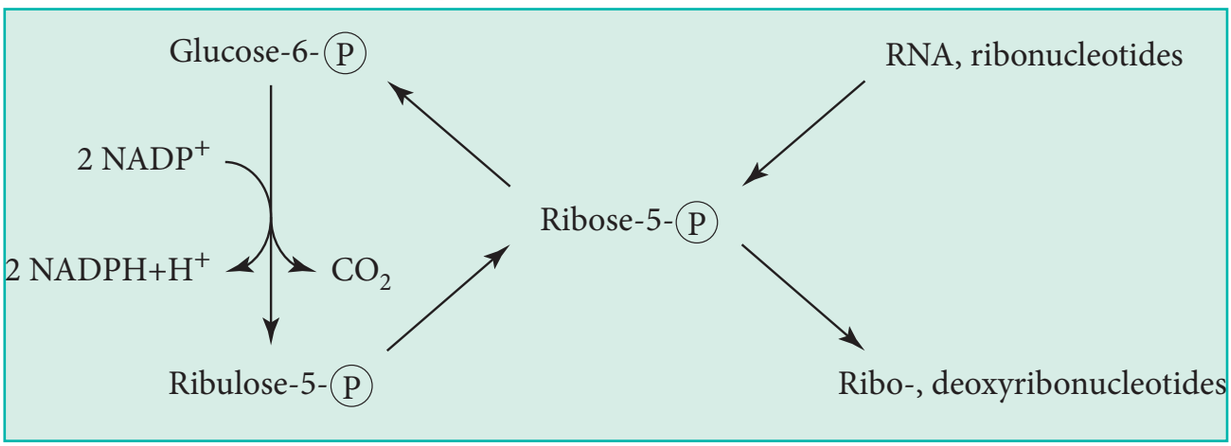
## 3.5. HEXOSE MONOPHOSPHATE SHUNT

So far we have learnt that glucose is oxidised by glycolysis & citric acid cycle, to generate energy and produce the necessary precursor metabolites. However, a number of alternative pathways have also been discovered. Hexose Monophosphate Shunt Pathway (HMP shunt). is one such important pathway. This occurs in the extra mitochondrial soluble portion of the cells.

This pathway, starts with glucose-6-phosphate as a key metabolite. Since, this pathway comprises both pentoses and hexoses, it is also referred to as the pentose phosphate pathway. This pathway consists of two phases namely the oxidative phase and the regenerative phase (non oxidative phase).

The oxidative phase, in which glucose-6-phosphate is oxidized and decarboxylated to give ribulose-5-phosphate. In this phase two NADPH (NADH equivalent) is formed. The fundamental difference between NADPH and NADH is that NADH is oxidised by the respiratory chain to generate ATP whereas NADPH serves as a hydrogen and electron donor in reductive biosynthesis of molecules such as fatty acids and cholesterol.

In cells that are not using ribose-5-phosphate for biosynthesis, the nonoxidative phase recycles six molecules of the pentose into five molecules of the hexose, glucose-6-phosphate, allowing continued production of NADPH and converting glucose-6-phosphate (in six cycles) to $CO_2$.






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